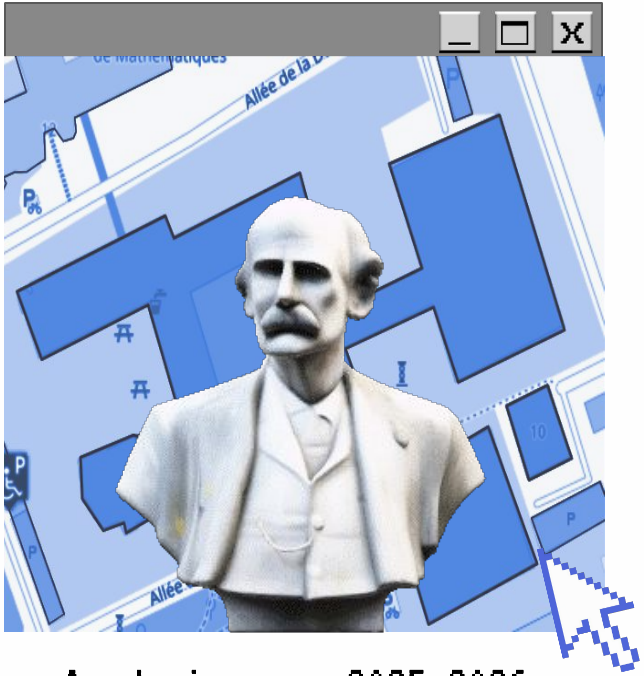
Monte Maps

ELEN0449 - Presentation



Academic year 2025-2026

Group 04:
Simon Gardier
Camille Trinh
Lei Yang

Motivation

Use case

Ever got students asking for directions ?
Difficulty to explain the path because of repetitive corridor ?

→ Need a tool to help people to navigate in a complex building

Solution

Monte-Maps
A real-time, on device, pipeline for visual localization in the Montefiore institute

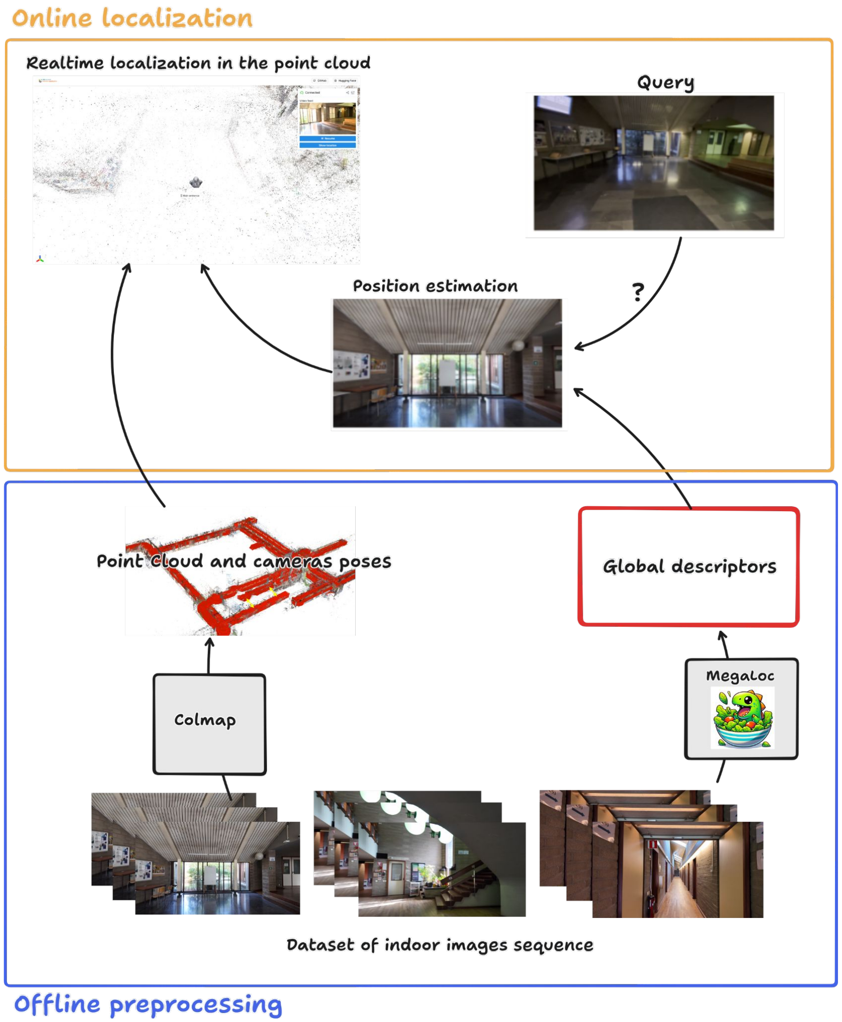
Pipeline

Task 1: 3D Reconstruction

- ❑ Given images $\{I_k\}$
- ❑ Predict $\{(x, y, z, r, g, b)_n\}$

Task 2: Position estimation

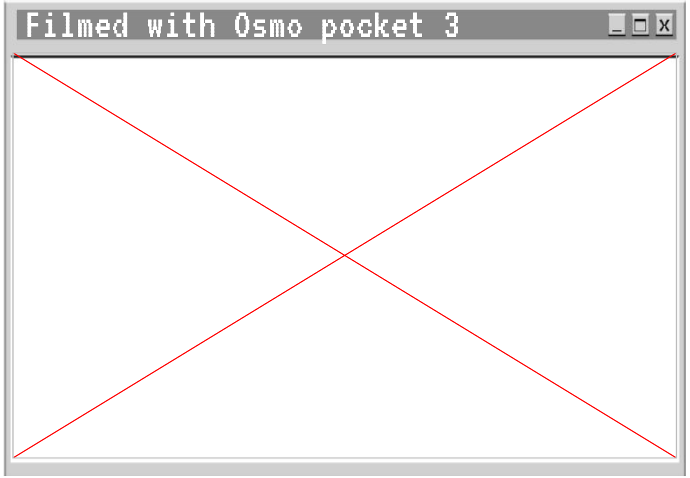
- ❑ Given $\{(x, y, z, r, g, b)_n\}$
- ❑ Given image $\{I\}$
- ❑ Predict $t \in R^{3x1}$



Data

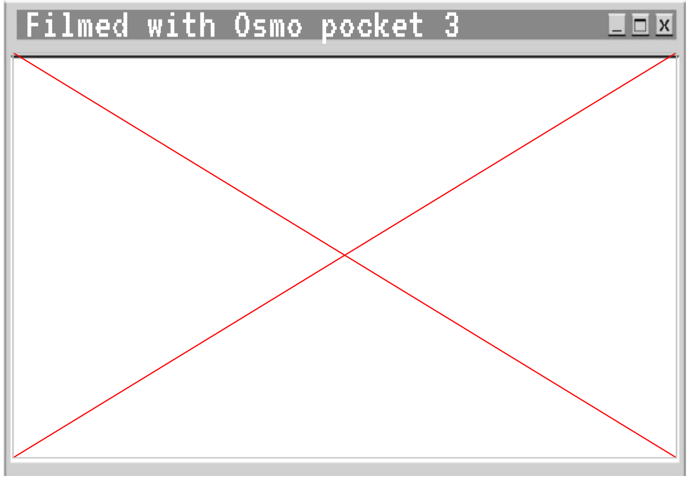
Data acquisition :

- Video capture of the building
- Each hallway is filmed in two directions of walking
- The video is sampled at 10 fps to create the input for COLMAP
- No obstructions or moving objects
- All lights are lit, combined with daylight



(Empty building)

Pipeline - Preprocessing

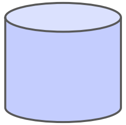


COLMAP

3D reconstruction with
camera pose estimation

MegaLoc

Descriptors database
→ 12,000 descriptors
→ Descriptors dimension = 8192



Pipeline - Position estimation (Retrieval)

MegaLoc

Query Image

Mega
Loc

Database of global
descriptors

Top-***1*** cosine similarity retrieval

Pipeline - Position estimation (refinement)

- Kalman filter
  - Output temporally smooth estimations
  - 1 Sensor: MegaLoc predicted images position,
  - Physical model: human walking at constant velocity
  - State: $[x, y, z, v_x, v_y, v_z]$

- Outliers rejection
  - Reject obvious outliers
  - Instead of a fixed Euclidean distance, use the filter uncertainty with Mahalanobis distance

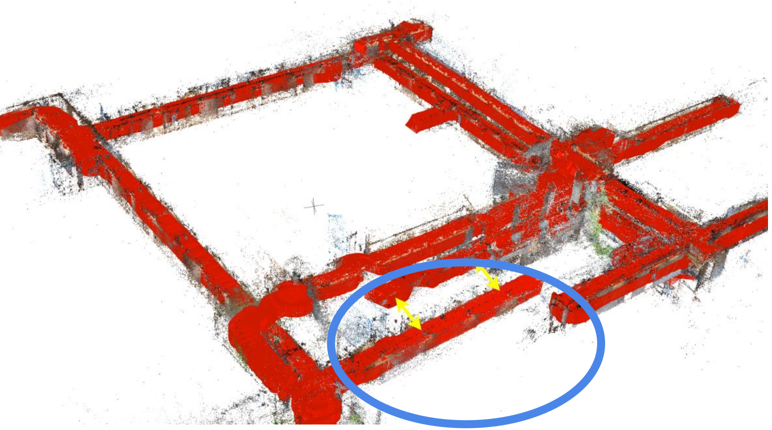
- Initialization phase
  - Capture positions during n seconds during which user should look around
  - Set initial position as median
  - Go back to initialization phase if no good measurement during *k* seconds

Evaluation - 3D Reconstruction

- No Ground Truth for evaluation → Only qualitative evaluation

Fails

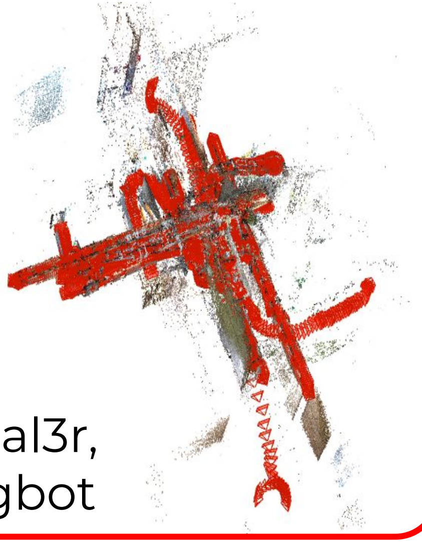
COLMAP
incremental mapper



Drift during
reconstruction

Depth-Anything-3

COLMAP
Global mapper

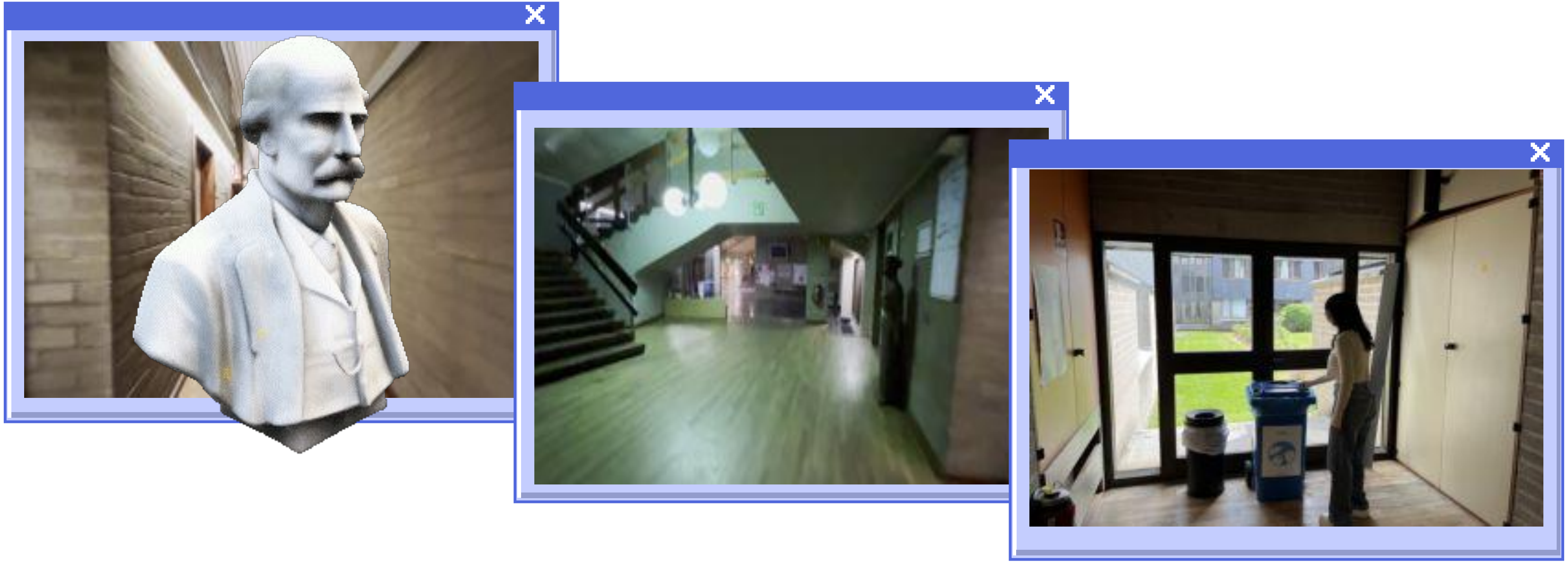


Other tested methods: Scal3r, VGGT-long, Pi3X-long, lingbot

Evaluation Data - Position estimation (Retrieval)

Small dataset of images to evaluate quantitatively and qualitatively the pipeline

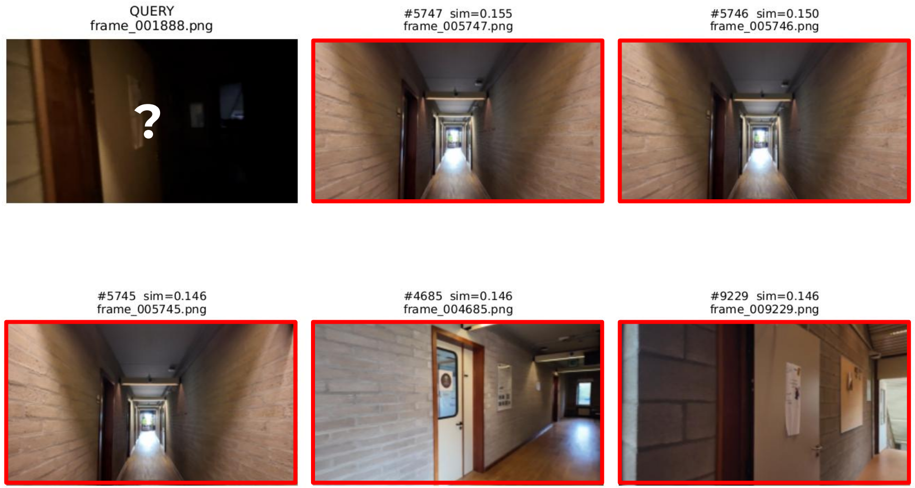
- Taken using a different camera than for capture (iPhone 13 Mini)
- Including night images with artificial lighting
- Including obstructions and objects that were moved
- 47 images, 2 videos



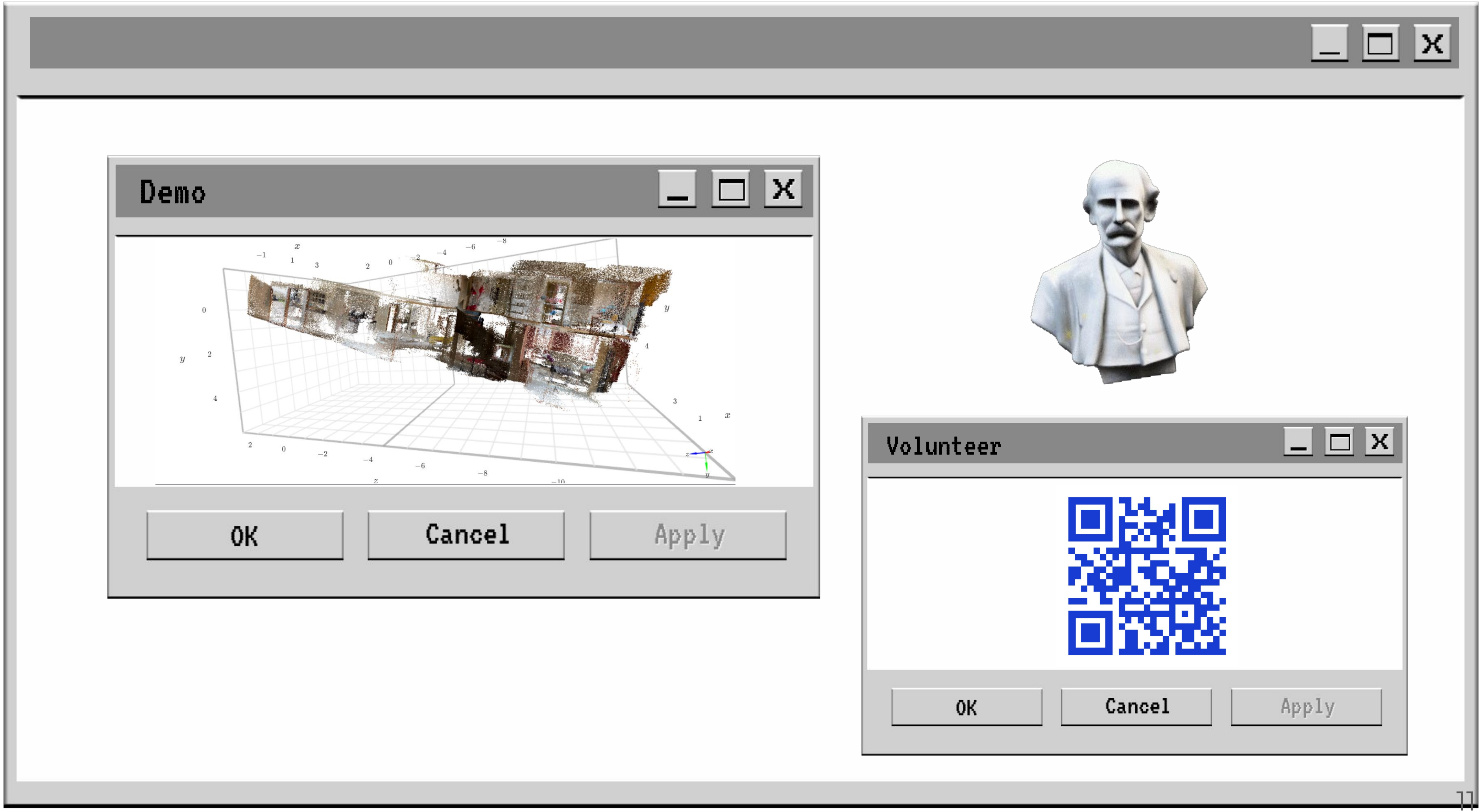
Evaluation - Position estimation (Retrieval)

Successful retrieval

Failed retrieval (very low light query)



- Retrieve top-***5*** images from the capture images
- Using the cosine similarity to compute the score: $\cos(\theta) = \dfrac{\mathbf{A} \cdot \mathbf{B}}{\|\mathbf{A}\|\|\mathbf{B}\|} = \dfrac{\sum\limits_{i=1}^{n} A_i B_i}{\sqrt{\sum\limits_{i=1}^{n} A_i^2} \sqrt{\sum\limits_{i=1}^{n} B_i^2}}$
- Successful retrieval 45/47 images (**95.74**%)


Demo
OK
Cancel
Apply
Volunteer
OK
Cancel
Apply

Conclusion

Aspects of Computer Vision explored

- 3D Reconstruction (COLMAP)
- Visual Recognition Pattern (MegaLoc)
- Indoor Localization

A dataset specific to Montefiore

- A video of Montefiore of 20 minutes
- A dataset to evaluate the pipeline

Use case

- Real-time, on device, pipeline for visual localization in Montefiore
- Possible extension (e.g., B37 with more videos)

Literature review, testing, and evaluation

- Optimization methods VS FF methods
- Tested Depth-Anything-3, VGGT-LONG, Pi3X-long, Scal3R, LingBot, and GOLMAP
- Find an appropriate metric for this pipeline and view the results

Q&A